Homework 6 - Class Question

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Writing a reusable function

Intro

The purpose of this assignment is to come up with a reusable R function that will help us analyze protein structures by using PDB data. To make the process more efficient so that we don't have to repeat the same code every time, the function will analyze the protein structure and generate an according plot. To test the function, I am using PDB Ids provided: 4AKE, 1AKE, 1E4Y

> Q6. Write your own function starting from the code above that analyzes protein drug interactions by reading in any protein PDB data and outputs a plot for the specified protein.

```
library(bio3d)
```

Creating Reusable Function

```
# Function: plot_protein
# Pdb_id is the PDB identifier for the protein structure.
# Default chain is "A" which specifies the protein chain we wan
# We are trying to analyze protein B values from the PDB structu


plot_protein <- function(pdb_id, chain = "A") {
  # reading the protein structure using its PDB ID
  protein<-read.pdb(pdb_id)

  # Selecting alpha carbon atoms from specified chain
  protein_ca<- trim.pdb(protein, chain = chain, elety="CA")

  # Taking out B-factor values from the specific atoms
  b_factors <-protein_ca$atom$b

  # Across all the proteins, plotting B-factor values
```

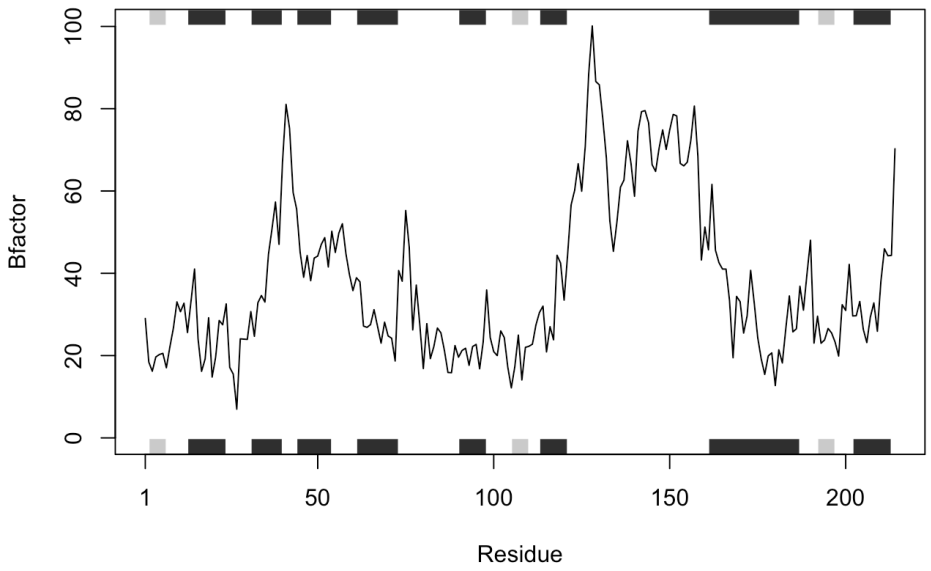
```
  plotb3(b_factors, sse = protein_ca, typ = "l",
         ylab="Bfactor")
}
```

Testing the Function with Outputs

```
# To test the function, I will still use the protein structures

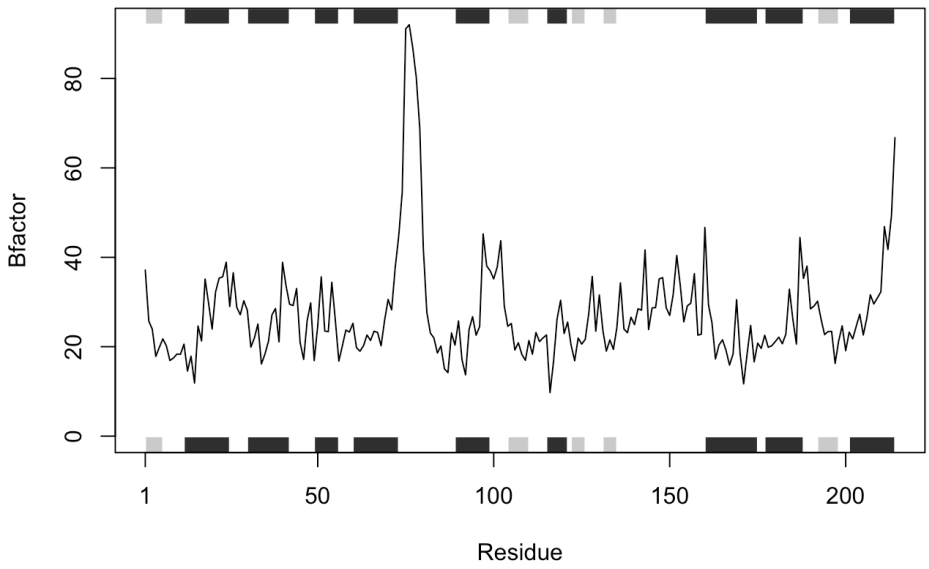
plot_protein("4AKE")
```

```
Note: Accessing on-line PDB file
```



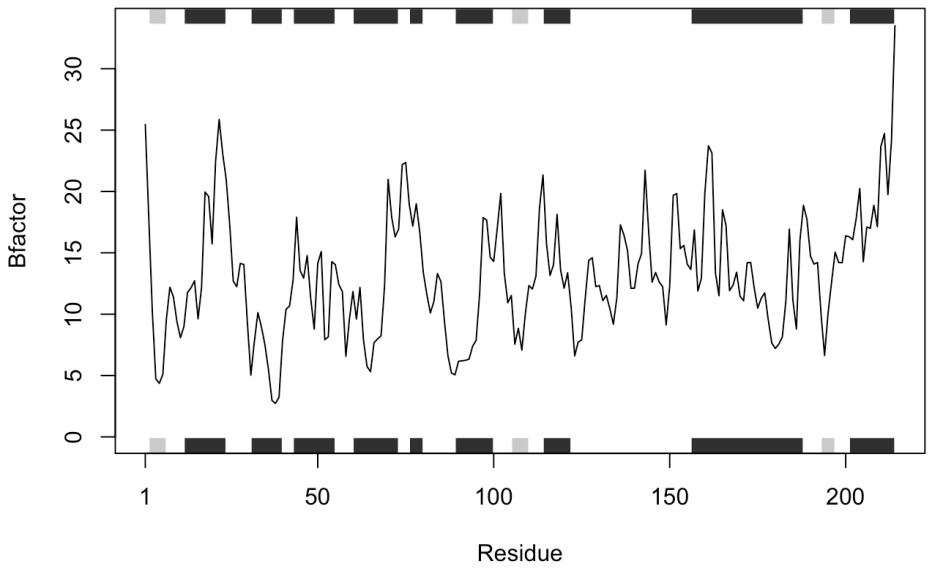

```
plot_protein("1AKE")
```

```
Note: Accessing on-line PDB file
 PDB has ALT records, taking A only, rm.alt=TRUE
```




```
plot_protein("1E4Y")
```

```
Note: Accessing on-line PDB file
```



Conclusion

The x-axis shows residue positions, while y-axis shows B-factor values. As an output we get a plot and not a table with data or numerical values. We can tell greater atonomic displacement by analyzing the higher B factor values.